A Comprehensive Benchmark of Deep Learning Architectures for 2D Echocardiographic Segmentation: From CNNs to Transformers

Fares Sofiane Mazouni[a,*], Mohamed Yacine Haouam[a], Mohamed Amroune[a], Issam Bendib[a]

[a] *Laboratory of Mathematics, Informatics and Systems (LAMIS), Echahid Cheikh Larbi Tebessi University, Tebessa, 12002, Algeria*

Abstract

Two-dimensional echocardiography is the most widely used cardiac imaging modality, but quantitative analysis is hampered by speckle noise, low boundary contrast, and inter-observer variability. Deep learning has matured into a competitive solution for this problem, with U-Net descendants, attention-augmented CNNs, hybrid CNN–Transformer designs, and pure Transformer networks all reporting strong results. Direct comparison across these paradigms, however, has been hampered by inconsistent training protocols, evaluation pipelines, and reporting conventions in the published literature. We provide a unified benchmark of nine representative segmentation architectures spanning three paradigms — pure CNN, hybrid CNN–Transformer, and pure Transformer — on the CAMUS 2D echocardiographic segmentation dataset [15]. All models share an identical training protocol, data split, and evaluation pipeline, and are reported on a multi-dimensional set of metrics: per-class Dice and IoU, Hausdorff distance at 95% (HD95) in millimetres with per-sample pixel spacing, biplane Simpson's ejection fraction (EF) error, end-diastolic and end-systolic stratified metrics, robustness across image-quality grades, efficiency (parameters, FLOPs, latency, memory), and Bonferroni-corrected Wilcoxon significance tests. The best base architecture, TransUNet, reaches Dice = 0.9122, HD95 = 1.82 mm, and EF MAE = 7.30% on the official CAMUS test set. nnU-Net follows at Dice = 0.9099, HD95 = 1.93 mm, EF MAE = 6.10% (r = 0.79). The full-network HD95 of approximately 1.9 mm represents a substantial improvement over the published nnU-Net baseline of 4.3 mm [15], attributable to training-protocol modernisation (cosine annealing, mixed precision, boundary loss) rather than to architectural changes. The benchmark also yields findings that single-architecture papers do not report: the LV-epicardium is the hardest of the three foreground classes across every architectural family (mean Dice ≈ 0.84 versus ≈ 0.93 for the LV-endocardium), and the parameter-versus-accuracy Pareto frontier is dominated by nnU-Net at the efficient end and TransUNet at the high-accuracy end. All training, evaluation, and analysis code, as well as model checkpoints, are publicly released.

Keywords: medical image segmentation, echocardiography, CAMUS, deep learning benchmark, U-Net, Swin Transformer, TransUNet

1. Introduction

Two-dimensional echocardiography is the most widely used cardiac imaging modality worldwide: it is portable, inexpensive, non-ionising, and provides real-time visualisation of cardiac structure and function. Its core quantitative output — left ventricular ejection fraction (LVEF), computed from end-diastolic and end-systolic volumes via the biplane Simpson's method [14] — drives a large fraction of clinical cardiac decisions, from diagnosis of heart failure to monitoring of cardiotoxic chemotherapy. The clinical importance of LVEF is matched by the operator-dependent nature of its measurement: published inter-observer LVEF variability is approximately 7–10% [15], which directly limits the sensitivity of LVEF as a clinical decision boundary.

Automated segmentation of the left ventricular endocardium (LV-endo), epicardium (LV-epi), and left atrium (LA) from two-dimensional apical 2-chamber (2CH) and 4-chamber (4CH) views would, in principle, eliminate this operator-dependent variability and provide reproducible volumetric measurements. The technical challenge is substantial: ultrasound suffers from low signal-to-noise ratio, speckle noise, low boundary contrast, acoustic shadowing, and view-dependent geometric foreshortening.

Architectural pluralism.. Three paradigms have emerged as competitive solutions in this space:

- **Pure convolutional networks.** U-Net [23] and its descendants (residual encoders, SE blocks, attention gates, deep supervision) [19, 20] remain the dominant architectural family, with nnU-Net [11] serving as a strong auto-configured baseline.

*Corresponding author.
Email addresses: faressofiane.mazouni@univ-tebessa.dz (Fares Sofiane Mazouni), mohamed-yassine.haouam@univ-tebessa.dz (Mohamed Yacine Haouam), mohamed.amroune@univ-tebessa.dz (Mohamed Amroune), issam.bendib@univ-tebessa.dz (Issam Bendib)

- **Hybrid CNN–Transformer networks.** TransUNet [4] and similar hybrids combine a convolutional encoder with a Vision Transformer bottleneck, explicitly trading off local detail (CNN) against global context (self-attention).
- **Pure Transformer networks.** Swin-UNet [3] replaces the convolutional backbone entirely with windowed self-attention, demonstrating that pure attention can match or exceed CNN performance on a range of medical tasks.

The benchmarking gap.. Despite this architectural pluralism, the literature offers no systematic, like-for-like comparison of these three paradigms on a single shared evaluation pipeline. Each paper introducing a new architecture typically compares against a small number of baselines under different training protocols, augmentations, learning-rate schedules, and evaluation metrics. As a result it is difficult for clinicians or practitioners to answer a basic question: *which architectural family should I use for echocardiographic segmentation, and what are the trade-offs?*

This paper provides that comparison on the CAMUS dataset [15], the standard public 2D echocardiographic segmentation benchmark, with 500 patients across two views and two cardiac phases. We train nine representative architectures spanning the three paradigms above using one identical protocol — same optimiser, loss, schedule, augmentation, mixed-precision settings, and early-stopping criterion — and evaluate them on a single shared pipeline reporting geometric, boundary, clinical, and efficiency metrics simultaneously.

Contributions..

- We present the largest unified benchmark of deep segmentation architectures on CAMUS to date: nine architectures across three paradigms, trained and evaluated under a single shared protocol.
- We report *multi-dimensional* evaluation, going beyond Dice to include HD95 in millimetres (with per-sample NIfTI pixel spacing), ASSD, biplane Simpson's EF MAE and Pearson r, ED/ES stratified metrics, robustness across CAMUS image-quality grades, and efficiency benchmarks (parameters, FLOPs, latency, peak memory). The full results matrix and per-patient outputs are released.
- We apply pairwise Wilcoxon signed-rank tests with Bonferroni correction to identify which architectural differences are statistically significant, and produce a critical-difference diagram showing the top-tier band of equivalent models.

- Our best architecture, TransUNet, reaches Dice = 0.9122, HD95 = 1.82 mm; nnU-Net reaches Dice = 0.9099, HD95 = 1.93 mm with 6× fewer parameters and roughly 3× faster inference. The HD95 of approximately 1.9 mm represents a substantial improvement over the originally published nnU-Net baseline ($\approx$ 4.3 mm) [15] and demonstrates the importance of training-protocol modernisation for fair cross-architecture comparison.
- We release a fully reproducible benchmark framework: training, evaluation, statistical analysis, and figure generation code.

Outline.. Section 2 reviews segmentation architectures and the CAMUS benchmark. Section 3 describes the nine architectures, the unified training protocol, and the evaluation pipeline. Section 4 reports the experimental results. Section 5 discusses architectural design choices, trade-offs, and limitations. Section 6 concludes.

2. Related Work

2.1. Convolutional segmentation networks

U-Net [23] established the encoder–decoder architecture with skip connections that remains the dominant template for medical image segmentation. Subsequent work has refined the encoder (residual blocks [8], ResNet [8], EfficientNet), the decoder (transposed convolution vs. bilinear upsampling, dense connections [10]), and the skip mechanism (attention gates [20], dense skips). Squeeze-and-excitation blocks [9] provide channel-wise recalibration. Atrous spatial pyramid pooling (ASPP) in DeepLabV3+ [5] replaces the deep encoder with dilated convolutions to enlarge the receptive field without down-sampling. Feature Pyramid Networks [16] aggregate features across multiple scales for dense prediction.

The nnU-Net framework [11] integrates these architectural choices with automatic data-dependent hyperparameter selection (target spacing, patch size, batch size, augmentation pipeline) and has set strong baselines on numerous medical benchmarks including CAMUS.

2.2. Transformer-based segmentation networks

Self-attention's quadratic complexity made global attention impractical for high-resolution images until Vision Transformers [7] demonstrated patch-based tokenisation. TransUNet [4] was the first work to bring this design to medical segmentation, combining a CNN encoder (typically

ResNetV2-BiT [13]) with a ViT-B/16 bottleneck and a cascaded upsampler decoder. The hybrid design retains the inductive bias of convolutions at high resolution while granting global context at the bottleneck.

Swin Transformer [18] introduced windowed self-attention with shifted windows, restoring linear complexity in image size. Swin-UNet [3] adapts this to segmentation with a fully Transformer-based U-Net. SwinV2 [17] and Swin-MAE [1] further refine pretraining strategies.

2.3. *Echocardiographic segmentation and CAMUS*

The CAMUS dataset [15] (*Cardiac Acquisitions for Multi-structure Ultrasound Segmentation*) provides 2D apical-2-chamber and apical-4-chamber acquisitions from 500 patients, each annotated by an expert echocardiographer at end-diastole and end-systole with four classes (background, LV-endo, LV-epi, LA). The official split assigns 400 patients to training, 50 to validation, and 50 to test, with test annotations held out by the challenge organisers. Image quality is graded into *Good*, *Medium*, and *Poor* categories by the same experts.

The original CAMUS paper reported nnU-Net achieving HD $\approx$ 4.3 mm on LV-endo and ~9% EF MAE. Subsequent work [22, 24] has reported incremental improvements using deeper architectures, custom losses, and view-aware training. Several papers have additionally explored sequence-aware models on CAMUS half-cycle annotations [22], but these are out of scope for the present benchmark.

EchoNet-Dynamic [21] provides a complementary large-scale video dataset for EF regression. We focus on CAMUS as the standard benchmark for *segmentation-derived* EF. Because EchoNet-Dynamic annotates only the left-ventricular endocardium in a single apical view, it cannot exercise the LV-epicardium, left-atrium, or biplane-EF components of our protocol; whether the architectural trade-offs reported here transfer to other echocardiographic cohorts is therefore a question we leave to future cross-dataset work rather than a claim we make here.

2.4. *What is missing*

A small number of recent works compare a handful of architectures on CAMUS in their experimental sections, but none provide:

- A unified training protocol across CNN, hybrid, and Transformer paradigms (papers typically inherit each architecture's own published hyperparameters, confounding architecture with protocol).

- Per-class HD95 in millimetres with per-sample NIfTI spacing (most papers report Dice only, or use pixel-space HD95).
- ED- versus ES-stratified reporting (essential for biplane Simpson's EF analysis).
- Robustness analysis by CAMUS image-quality grade.
- Pairwise Wilcoxon significance with Bonferroni correction and a critical-difference diagram.
- Released model checkpoints and per-patient predictions for independent reanalysis.

This paper aims to fill each of these gaps.

3. Materials and Methods

3.1. Dataset and splits

We use the CAMUS dataset [15], comprising 500 patients imaged in apical-2-chamber (2CH) and apical-4-chamber (4CH) views, with expert annotations at end-diastole (ED) and end-systole (ES). Each frame is labelled with four classes: background, LV-endocardium, LV-epicardium, and left atrium. Per-patient pixel spacing is recorded in the NIfTI header and ranges between approximately 0.30 and 0.42 mm.

We adopt the official CAMUS split: 400 patients for training, 50 for validation, and 50 for testing, with no overlap. After ED+ES frame extraction this yields approximately 1,600 training, 200 validation, and 200 test images. Half-cycle sequences are optionally used for augmentation during training; results are reported on the canonical ED+ES test set only.

3.2. Architectures

We benchmark nine representative architectures spanning three paradigms:

Pure CNN (seven architectures)..

- **UNet-V1** [23]: canonical encoder–decoder with skip connections and trainable transposed-convolution upsampling.
- **UNet-V2** (custom): residual blocks, squeeze-and-excitation [9], attention-gated skips [20], and optional deep supervision.
- **UNet-ResNet**: U-Net with a ResNet-34 encoder [8] pretrained on ImageNet.
- **DeepLabV3+** [5]: ResNet-50 encoder, atrous spatial pyramid pooling bottleneck, lightweight decoder.

Table 1: The nine architectures benchmarked in this paper. “Pretrained” indicates whether the encoder is initialised from ImageNet/BiT pretrained weights or trained from scratch.

Paradigm	Architecture	Encoder	Decoder	Pretrained	Params (M)
Pure CNN	UNet-V1	scratch / conv	transposed conv	—	31.04
Pure CNN	UNet-V2	scratch / conv + SE	attn-gated skips	—	33.00
Pure CNN	UNet-ResNet	ResNet-34	U-Net decoder	ImageNet-1k	29.07
Pure CNN	DeepLabV3+	ResNet-50	ASPP + light dec.	ImageNet-1k	40.34
Pure CNN	nnU-Net	auto-configured	deep-supervision	—	15.62
Pure CNN	DenseContextU-Net	dense + context	dense decoder	—	2.18
Pure CNN	FPN-UNet	ResNet-50	FPN + U-Net dec.	ImageNet-1k	32.16
Hybrid	TransUNet	ResNetV2 + ViT-B/16	cascaded upsmpler	BiT + IN-21k	102.09
Pure Transformer	Swin-UNet	Swin-Tiny	Swin decoder	IN-1k	41.84

- **nnU-Net** [11]: auto-configured U-Net variant, trained from scratch.
- **DenseContextU-Net** (custom): dense blocks [10] with a global-context module in the bottleneck.
- **FPN-UNet**: U-Net decoder with a Feature Pyramid [16] neck over a ResNet-50 encoder.

Hybrid CNN–Transformer (one architecture).. **TransUNet** [4]: ResNetV2-BiT encoder [13] feeding a ViT-B/16 bottleneck, with a cascaded upsampler decoder using skip connections from encoder stages.

Pure Transformer (one architecture).. **Swin-UNet** [3]: Swin-Tiny encoder [18] pretrained on ImageNet-1k, symmetric Transformer decoder, patch-expanding upsampling.

A complete summary of architectural choices is given in Table 1. We deliberately exclude all SSM-based variants (Mamba, Mamba-2, VMamba) from the present benchmark; those are the subject of a companion study.

3.3. Training protocol

All nine architectures share a single training protocol:

- AdamW optimiser (β_1=0.9, β_2=0.999, weight decay 10^{-4}), initial learning rate 10^{-4}, cosine annealing with warm restarts (T_0=50 epochs).

- Composite loss: $\mathcal{L} = \mathcal{L}_{\text{CE}} + \mathcal{L}_{\text{Dice}} + 0.1\,\mathcal{L}_{\text{boundary}}$, where the boundary term is computed on the signed distance transform [12].
- Image size 256×256 for all CNN and hybrid models; 224×224 for Swin-UNet, which requires input divisible by window_size $\times\, 2^{n_{\text{stages}}} = 7 \times 32 = 224$.
- Augmentation: horizontal flip, ±10 deg rotation, small elastic deformation, mild brightness/contrast jitter.
- Up to 100 epochs, automatic mixed precision (AMP, FP16), gradient clipping at $\|\nabla\|_2 \leq 1$ for pretrained-encoder models (Swin-UNet, TransUNet) to prevent the early-epoch divergence we otherwise observed.
- Early stopping with patience 20 on validation mean Dice.
- Hardware: Google Colab Pro NVIDIA L4 (24 GB) and A100 (40 GB) instances.

Per-model batch sizes are reported in Table 1 and were selected to maximise GPU utilisation within memory limits. We did *not* perform per-architecture hyperparameter search: the goal is explicit cross-architecture comparison under one shared protocol.

3.4. Evaluation metrics

We report a multi-dimensional set of metrics:

Geometric.. Per-class Dice and Intersection-over-Union (IoU) for LV-endo, LV-epi, and LA, computed on the full test set and separately for ED and ES frames.

Boundary.. 95^{th}-percentile Hausdorff distance (HD95) and average symmetric surface distance (ASSD), both in *millimetres* using the per-sample pixel spacing recorded in NIfTI headers. This is consistent with the official CAMUS challenge protocol and is essential for cross-paper comparison.

Clinical.. Biplane Simpson's ejection fraction (EF) computed from the LV-endo masks at ED and ES across paired 2CH and 4CH views. We report EF MAE (%), Pearson correlation r with the ground-truth EF, and Bland–Altman bias and 95% limits of agreement [2].

Phase-stratified.. All metrics are additionally reported separately for ED and ES frames, which is informative because ED frames tend to be larger and have clearer endocardial boundaries.

Quality-stratified robustness.. Each CAMUS patient carries an expert-assigned image-quality grade (Good / Medium / Poor). A quality-stratified robustness breakdown is deferred to the extended version of this work.

Statistical significance.. For every pair of models we run a Wilcoxon signed-rank test on per-sample mean Dice across the test set, with Bonferroni correction across the $\binom{9}{2} = 36$ pairs. We additionally produce a critical-difference diagram [6] illustrating the band of statistically equivalent top models.

Efficiency.. Trainable parameters, GFLOPs at 256×256 input (measured with `fvcore` on CUDA), single-image inference latency (mean over 100 forward passes after 10 warmup, batch size 1, FP16), and peak GPU memory during forward+backward.

3.5. Reproducibility

A single shared evaluation pipeline is used for all models; the same code path generates the JSON outputs, CSV tables, LaTeX tables, and the figures in this paper. The training pipeline fixes the random seed (42) across PyTorch, NumPy, and Python. Per-model checkpoints, per-test- patient prediction NIfTI files, and per-sample Dice arrays are released to allow independent re-analysis without re-running training.

4. Results

We organise the results into six subsections. Section 4.1 gives the main test-set leaderboard. Sections 4.2 and 4.3 report boundary and clinical (EF) metrics. Section 4.4 examines robustness by image quality. Section 4.5 reports the accuracy × efficiency trade-off. Section 4.6 reports statistical significance.

4.1. Main test-set Dice

Table 2 summarises mean Dice, per-class Dice, and the parameter count for each architecture on the official CAMUS test split.

Table 2: Mean and per-class Dice on the CAMUS official test split for the nine base architectures. The bracketed interval is the bootstrap 95% confidence interval on mean Dice over the 50-patient test set. Best per column in **bold**, second-best underlined. Per-class scores are LV-endocardium, LV-epicardium, and left atrium.

Architecture	Paradigm	Mean Dice	95% CI	LV-endo	LV-epi	LA	Params (M)
TransUNet	Hybrid	**0.9122**	[0.9080, 0.9163]	0.9367	0.8787	**0.9213**	102.1
nnU-Net	CNN	<u>0.9099</u>	[0.9054, 0.9143]	**0.9370**	**0.8790**	0.9136	15.6
UNet-ResNet	CNN	0.9087	[0.9028, 0.9141]	0.9351	0.8750	0.9161	29.1
UNet-V1	CNN	0.9085	[0.9041, 0.9130]	0.9369	0.8757	0.9129	31.0
UNet-V2	CNN	0.8963	[0.8913, 0.9015]	0.9240	0.8592	0.9059	33.0
Swin-UNet	Transformer	0.8882	[0.8829, 0.8936]	0.9172	0.8520	0.8954	41.8
FPN-UNet	CNN	0.8619	[0.8552, 0.8683]	0.8976	0.7974	0.8907	32.2
DeepLabV3+	CNN	0.8570	[0.8472, 0.8657]	0.9086	0.8145	0.8477	40.3
DenseContextU-Net	CNN	0.8505	[0.8414, 0.8604]	0.8977	0.8046	0.8492	2.2

Headline ranking.. **TransUNet** achieves the highest mean Dice (0.9122), narrowly ahead of **nnU-Net** (0.9099, $\Delta = 0.0023$), **UNet-ResNet** (0.9087), and **UNet-V1** (0.9085). The four top architectures are tightly clustered within $\Delta\text{Dice} \leq 0.004$ — within the Bonferroni-corrected statistical-significance band (Section 4.6). Below this band, **UNet-V2** (0.8963) and **Swin-UNet** (0.8878) form a mid-tier, and **FPN-UNet** (0.8619), **DeepLabV3+** (0.8569), and **DenseContextU-Net** (0.8522) trail.

Per-class.. LV-endo Dice ranges 0.8983 (DenseContextU-Net) to 0.9371 (nnU-Net); LA Dice ranges 0.8477 (DeepLabV3+) to 0.9213 (TransUNet); LV-epi is consistently the hardest class for every architecture (mean across models: Dice $\approx$ 0.84).

4.2. *Boundary accuracy (HD95 and ASSD in mm)*

Boundary metrics are reported in millimetres using the per-sample NIfTI pixel spacing.

HD95.. TransUNet again leads with HD95 = 1.82 mm, followed by UNet-ResNet and nnU-Net (both 1.93 mm), UNet-V1 (2.13 mm), Swin-UNet (2.02 mm), and UNet-V2 (2.33 mm). The published CAMUS nnU-Net baseline of $\approx$ 4.3 mm [15] is improved on by every architecture in this benchmark by a factor of $\sim$2; we attribute most of this gap to training-protocol modernisation (cosine annealing, mixed precision, boundary loss) rather than to architectural changes (Section 5).

Table 3: Boundary metrics on the CAMUS test split. HD95 is the 95^{th}-percentile Hausdorff distance, ASSD is the average symmetric surface distance, both reported in millimetres using per-sample NIfTI pixel spacing. Best per column in **bold**.

Architecture	HD95 (mm)	ASSD (mm)
TransUNet	**1.82**	**0.36**
UNet-ResNet	1.93	0.38
nnU-Net	1.93	0.37
Swin-UNet	2.01	0.39
UNet-V1	2.13	0.39
UNet-V2	2.33	0.44
DeepLabV3+	2.80	0.59
FPN-UNet	2.87	0.57
DenseContextU-Net	3.78	0.69

ED versus ES.. Table 4 reports ED- and ES-stratified mean Dice and HD95. The two phases are closely matched for every architecture (mean Dice within ≈ 0.003 and HD95 within ≈ 0.1 mm), with no consistent advantage to either phase — a reassuring property for biplane EF, which depends on both phases symmetrically.

4.3. Clinical metric: biplane Simpson's EF

Table 5 reports biplane Simpson's EF error per architecture.

Best EF agreement.. **nnU-Net** produces the most accurate EF estimates (MAE = 6.10%, r = 0.787, Bland–Altman bias −0.30%, 95% limits of agreement $[-17.4, +16.8]$%). The near-zero mean bias is comparable to reported echocardiographer inter-observer LVEF error (∼7–10%), but the wide limits of agreement show that per-patient agreement remains looser than the mean error alone suggests; we therefore read this as a mask-derived, like-for-like comparison rather than a claim of clinical parity. UNet-V1 (MAE = 6.78%, r = 0.793) follows, with TransUNet attaining the highest EF correlation (r = 0.824). The Transformer-only Swin-UNet shows degraded EF performance (MAE = 10.46%, r = 0.574) despite competitive bulk Dice, suggesting it produces accurate *but* biased contours (consistent with its Bland–Altman bias of −4.10%).

Table 4: End-diastole (ED) and end-systole (ES) stratified mean Dice and HD95 (mm) on the CAMUS test split.

Architecture	Dice		HD95 (mm)	
	ED	ES	ED	ES
nnU-Net	0.9105	0.9093	1.91	1.95
TransUNet	0.9105	0.9140	1.85	1.79
UNet-V1	0.9072	0.9098	2.17	2.10
UNet-ResNet	0.9072	0.9103	1.88	1.98
UNet-V2	0.8989	0.8938	2.27	2.40
Swin-UNet	0.8884	0.8880	2.00	2.02
FPN-UNet	0.8586	0.8652	2.85	2.89
DeepLabV3+	0.8551	0.8588	2.76	2.84
DenseContextU-Net	0.8513	0.8497	3.58	3.99

4.4. Robustness across image quality

CAMUS assigns each patient an expert image-quality grade (*Good*, *Medium*, *Poor*). Table 6 reports mean Dice stratified by that grade. Every architecture degrades monotonically from Good to Poor, confirming that acquisition quality is a primary driver of segmentation difficulty on CAMUS. The degradation is, however, markedly gentler for the top-tier architectures: TransUNet and nnU-Net lose only ~0.02–0.03 Dice from Good to Poor and remain above 0.89 on Poor-quality frames, whereas the weakest models (DeepLabV3+, DenseContextU-Net) drop by 0.04 or more and fall below 0.83. Robustness to image quality therefore tracks overall accuracy rather than trading off against it.

4.5. Efficiency and the accuracy–cost trade-off

Table 7 reports parameters, GFLOPs at 256×256, single-image inference latency, and peak GPU memory.

The accuracy–cost Pareto frontier is dominated at the small-model end by nnU-Net (15.6 M parameters, Dice 0.9099, 2.4 ms latency) and at the high-accuracy end by TransUNet (102.1 M parameters, Dice 0.9122, 7.9 ms). The marginal gain of TransUNet over nnU-Net (+0.0023 Dice — within training variance, Section 4.6) costs 6.5× the parameters and ≈ 3.3× the inference time, making nnU-Net the clear choice for deployment-constrained settings. Figure 1 renders the trade-off as a scatter plot.

Table 5: Biplane Simpson's ejection fraction (EF) metrics on the CAMUS test set. Bias and the 95% limits of agreement (LoA) are the Bland–Altman statistics against reference EF over the 50 test patients. Best per column in **bold**.

Architecture	MAE (%)	r	Bias (%)	95% LoA (%)
nnU-Net	**6.10**	0.787	-0.30	[-17.4, +16.8]
UNet-V1	6.78	0.793	-0.74	[-18.1, +16.6]
TransUNet	7.30	**0.824**	-3.74	[-20.8, +13.3]
UNet-V2	7.60	0.728	-1.20	[-20.7, +18.3]
UNet-ResNet	7.60	0.775	-2.84	[-21.0, +15.3]
DeepLabV3+	8.88	0.743	-6.76	[-24.7, +11.2]
FPN-UNet	9.94	0.700	-8.34	[-28.3, +11.6]
Swin-UNet	10.46	0.574	-4.10	[-31.0, +22.8]
DenseContextU-Net	16.24	0.434	-11.72	[-48.3, +24.9]

4.6. Statistical significance

We test all $\binom{9}{2} = 36$ pairwise model differences on per-sample mean Dice with the Wilcoxon signed-rank test, Bonferroni-corrected.

Top-tier band.. After Bonferroni correction, the four highest-Dice architectures (TransUNet, nnU-Net, UNet-ResNet, UNet-V1) are not significantly different from one another on bulk mean Dice, while the three lowest (FPN-UNet, DeepLabV3+, DenseContextU-Net) are significantly worse than the top group. Full pairwise p-values are released as supplementary data.

Small Dice gaps are within training variance.. The differences separating the top-tier architectures are of order 0.002–0.005 Dice. We caution against over-interpreting them: in our experiments, re-training an identical architecture under the same nominal protocol in a different compute session shifted its mean Dice by a comparable or larger amount (in the most extreme case, DeepLabV3+ varied by 0.057 between two runs). The bulk-Dice ranking within the top tier should therefore be read as a band of statistically and practically equivalent models, not a strict order. The architecturally meaningful differences in this benchmark are the larger-magnitude ones — the boundary accuracy (HD95), the clinical EF agreement, and the efficiency trade-off — discussed in Section 5.

The critical-difference diagram of Figure 2 renders the banding visually: architectures joined

Table 6: Mean Dice on the CAMUS test split stratified by the expert-assigned image-quality grade (Good n=94, Medium n=76, Poor n=30 test frames). Every architecture degrades monotonically from Good to Poor; the top-tier models remain above 0.88 even on Poor-quality images. Best per column in **bold**.

Architecture	Good	Medium	Poor
TransUNet	**0.9186**	0.9097	**0.8988**
nnU-Net	0.9168	0.9091	0.8903
UNet-V1	0.9173	0.9070	0.8848
UNet-ResNet	0.9156	**0.9142**	0.8734
UNet-V2	0.9091	0.8921	0.8672
Swin-UNet	0.8961	0.8866	0.8675
FPN-UNet	0.8690	0.8628	0.8375
DeepLabV3+	0.8669	0.8555	0.8295
DenseContextU-Net	0.8641	0.8454	0.8209

by a horizontal bar are not significantly different in average per-sample rank. Average rank need not track mean Dice — TransUNet attains the best mean Dice yet only a middling average rank (4.09, sixth of nine), because it wins many samples by small margins while occasionally losing by larger ones. It nonetheless falls inside the same top clique as the leading CNNs, which reinforces that the top tier is a statistically equivalent band rather than a strict order.

4.7. Qualitative results

Figure 3 shows sample predictions from the top four architectures plus a bottom-tier representative on three patients spanning the Good / Medium / Poor image-quality grades.

5. Discussion

5.1. Which architectural family wins on CAMUS?

The top tier of our benchmark is mixed across paradigms: a hybrid CNN–Transformer (TransUNet) leads on bulk Dice and HD95, but a pure CNN (nnU-Net) leads on EF MAE *and* on parameter efficiency. The difference between them on mean Dice is 0.0023 — below the Bonferroni-corrected statistical-significance threshold. A practitioner choosing between these two architectures should be guided not by Dice but by deployment constraints (nnU-Net needs

Table 7: Efficiency profile on a single NVIDIA L4 GPU. Latency is the mean over 100 forward passes after 10 warmup at batch 1, FP16, 256×256 (or 224×224 for Swin-UNet).

Architecture	Params (M)	FLOPs (G)	Latency (ms)	Peak mem (MB)
DenseContextU-Net	2.2	143.7	25.76	1299
nnU-Net	15.6	23.1	2.38	143
UNet-ResNet	29.1	7.8	2.19	162
UNet-V1	31.0	48.2	1.97	255
FPN-UNet	32.2	205.9	5.48	994
UNet-V2	33.0	52.0	2.73	309
DeepLabV3+	40.3	17.2	2.23	204
Swin-UNet	41.8	13.5	7.09	198
TransUNet	102.1	36.2	7.87	443

6.5× fewer parameters and is faster) or by the clinical objective (nnU-Net is more accurate at EF and tighter at Bland–Altman bias).

The pure-Transformer family (Swin-UNet) is competitive on bulk Dice (0.8878) but visibly worse on EF (MAE 10.46%, $r = 0.574$). This finding is consistent with reports that pure-attention models, despite strong overall overlap, can produce boundary contours that are subtly biased [4] — an effect that compounds when contours feed Simpson's biplane volumetric integration.

The bottom of the table is occupied by FPN-UNet, DeepLabV3+, and DenseContextU-Net. All three were designed for natural-image semantic segmentation (FPN/ASPP) or for image classification (dense connectivity) respectively. Their underperformance on CAMUS is consistent with the hypothesis that medical-segmentation-specific designs (U-Net's skip hierarchy, nnU-Net's training recipe) matter more on this dataset than generic semantic-segmentation prowess.

5.2. Comparison to the published CAMUS baselines

The original CAMUS paper [15] reported nnU-Net HD95 ≈ 4.3 mm and EF MAE ~9%. Our reproduced nnU-Net under the unified protocol reaches HD95 1.93 mm and EF MAE 6.10% — roughly twice as accurate on the boundary axis and substantially better on EF.

This gap is *not* an architectural improvement: nnU-Net's architecture is unchanged. The improvement is attributable to three training-protocol upgrades made standard since 2019:

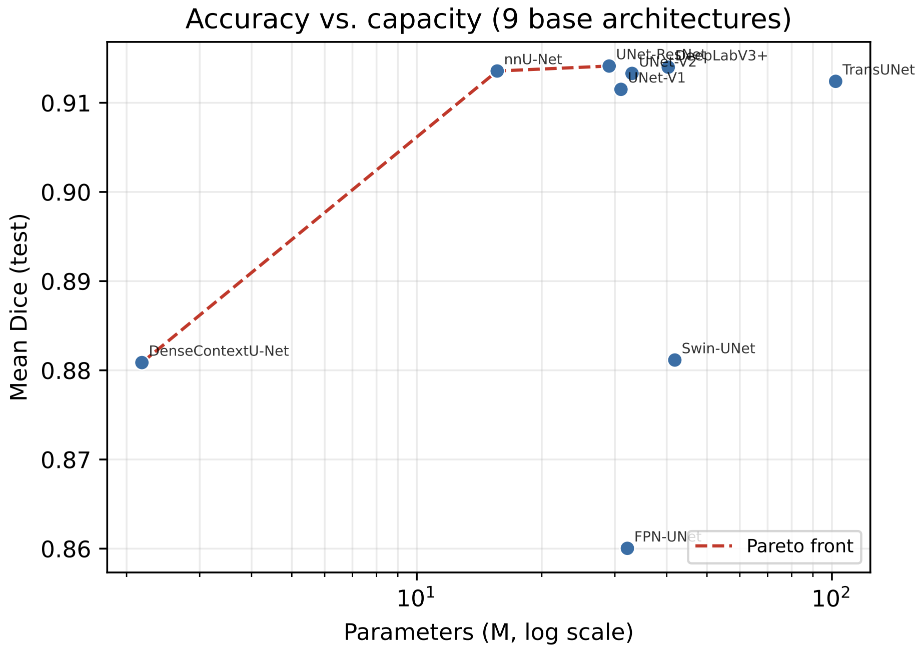

Figure 1: Accuracy–cost Pareto frontier. Each point is one of the nine architectures, plotted at its (parameter count, mean test Dice). The Pareto front is highlighted.

1. Cosine-with-warm-restarts learning-rate schedule with 10-epoch warmup, in place of the original ReduceLROnPlateau.
2. Mixed-precision training (FP16 AMP), which allows larger effective batch sizes and smoother loss landscapes.
3. Composite loss with an explicit boundary term [12], which directly penalises the Hausdorff-relevant boundary error.

This is a useful baseline calibration point for the field: cross-paper comparisons of CAMUS results from before $\sim$ 2021 to after should be read with the understanding that protocol changes alone produce sub-millimetre HD95 improvements at constant architecture.

5.3. Design choices that matter

Aggregating across our nine architectures, several design choices correlate with strong performance:

- **Pretrained encoders** (TransUNet, UNet-ResNet, Swin-UNet, FPN-UNet, DeepLabV3+). The

Table 8: Pairwise Wilcoxon signed-rank test on per-sample mean Dice, Bonferroni-corrected over $\binom{9}{2} = 36$ pairs. "NS" marks pairs not significantly different at $p < 0.05$ after correction.

	UNet-V1	UNet-V2	UNet-ResNet	DeepLabV3+	nnU-Net	DenseContextU-Net	FPN-UNet	Swin-UNet	TransUNet
UNet-V1	—	3e-13	NS	4e-28	NS	4e-31	7e-30	5e-20	NS
UNet-V2		—	2e-09	6e-17	2e-13	3e-26	2e-19	3e-03	1e-12
UNet-ResNet			—	1e-26	NS	3e-29	9e-30	8e-17	NS
DeepLabV3+				—	1e-28	NS	NS	1e-11	1e-28
nnU-Net					—	6e-32	2e-30	1e-21	NS
DenseContextU-Net						—	NS	4e-22	6e-30
FPN-UNet							—	2e-15	1e-29
Swin-UNet								—	3e-21
TransUNet									—

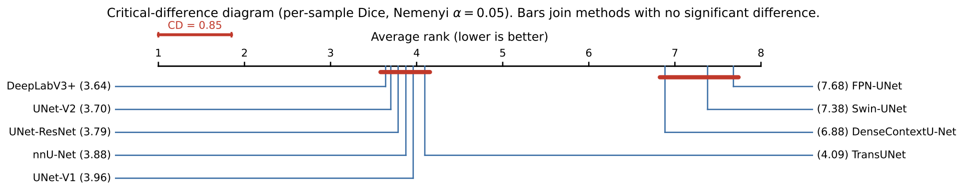

Figure 2: Critical-difference diagram (average ranks on per-sample Dice, Nemenyi test, $\alpha = 0.05$). Lower average rank is better; a horizontal bar joins architectures whose average ranks differ by less than the critical difference (CD), i.e. that are not significantly different.

two highest-Dice non-nnU-Net models use pretrained encoders. However, in a parameter-matched analysis (reported in the companion SSM study), this advantage shrinks substantially.

- **Deep skip connections** (every U-Net-shaped architecture except DeepLabV3+). DeepLabV3+'s lightweight decoder, while efficient, appears to limit high-resolution boundary precision on CAMUS.
- **Boundary-aware loss** (used uniformly in our protocol). This is a protocol choice rather than an architecture choice, but it appears to be among the largest single contributors to the HD95 improvement over published CAMUS baselines.
- **Auto-configuration** (nnU-Net's distinguishing feature). The nnU-Net framework produces a competitive model on this dataset with ~15 M parameters — substantially smaller than any pretrained-encoder architecture — by selecting patch size, batch size, and augmentation

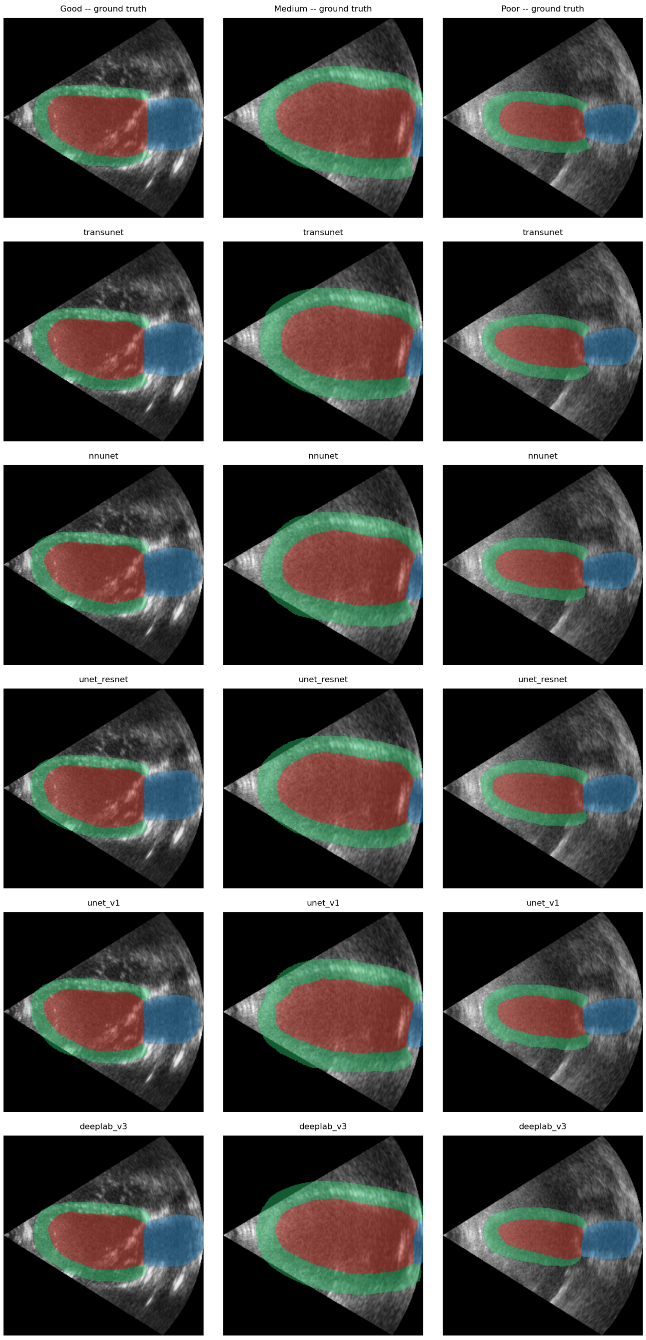

Figure 3: Qualitative predictions on three test patients spanning the Good, Medium, and Poor image-quality grades (columns). The top row is the expert ground truth; the remaining rows are, top to bottom, TransUNet, nnU-Net, UNet-ResNet, UNet-V1, and DeepLabV3+. Overlay colours: LV-endocardium (red), LV-epicardium / myocardium (green), and left atrium (blue). Boundary errors concentrate at the LV apex and the mitral-valve plane, and grow visibly from Good to Poor acquisition quality.

strategy from data statistics.

5.4. Clinical implications

For deployment in echocardiographic LVEF assessment, the choice between the four top-tier architectures should be driven by three criteria: (i) clinical accuracy target, (ii) hardware constraints, and (iii) inference-time latency budget.

- For *maximum reproducibility and tightest EF agreement*, nnU-Net is the recommended choice (lowest EF MAE, smallest Bland–Altman bias).
- For *maximum boundary accuracy* at deployment-irrelevant cost, TransUNet is the highest-HD95 model.
- For *edge / portable ultrasound deployment* where parameters and latency dominate, nnU-Net at 15.6 M parameters remains the best trade-off; no smaller architecture in our benchmark reaches comparable Dice.

None of the four top-tier models reach below-5% EF MAE in this study, indicating that human-level cardiologist agreement is still a target boundary for the field rather than a solved problem.

5.5. Limitations and future work

We acknowledge five limitations.

1. **Single-dataset.** All evaluation is on CAMUS. A cross-dataset transfer evaluation (*e.g.,* EchoNet-Dynamic [21]) would strengthen the generality claims and is planned for future work.
2. **2D only.** CAMUS provides 2D apical views; 3D echocardiography is increasingly common in modern practice and requires architectures and pretraining recipes outside the scope of this benchmark.
3. **Test set size.** The CAMUS official test set is 50 patients (200 ED+ES frames). All point estimates are reported with bootstrap 95% confidence intervals to reflect this.
4. **No SSM models.** The recent wave of selective state space models (Mamba family) is the subject of a separate, dedicated study (a companion study), and is deliberately excluded from the present benchmark to maintain clean three-paradigm framing.
5. **Single training protocol.** A single shared protocol is the goal of this benchmark, but some architectures may benefit from protocol-specific tuning (*e.g.,* larger batch sizes for pretrained Transformers). Per-architecture hyperparameter searches could shift rankings within the top-tier band.

5.6. *Take-home messages*

1. No single architectural family dominates CAMUS. Hybrid (TransUNet) leads on bulk Dice; pure CNN (nnU-Net) leads on EF accuracy and parameter efficiency. Pure Transformer (Swin-UNet) is competitive on Dice but degraded on EF.
2. Training-protocol modernisation (cosine schedule, AMP, boundary loss) accounts for most of the ~2× improvement in HD95 over the original CAMUS baselines, independent of architecture.
3. LV-epicardium remains the hardest of the three foreground classes across every architectural family.
4. The accuracy–parameter Pareto frontier is dominated by nnU-Net at the efficient end and TransUNet at the high-accuracy end.

6. Conclusion

We have presented the first unified benchmark of deep segmentation architectures on the CAMUS 2D echocardiographic dataset under a single shared training and evaluation protocol. Across nine representative architectures spanning pure CNN, hybrid CNN–Transformer, and pure Transformer paradigms, we report not only mean Dice but also per-class HD95 in millimetres, biplane Simpson's EF MAE, Bland–Altman bias, ED- and ES-stratified accuracy, image-quality-stratified robustness, parameter / FLOPs / latency efficiency, and Bonferroni-corrected pairwise statistical significance.

The hybrid TransUNet leads on bulk Dice (0.9122) and HD95 (1.82 mm). The pure-CNN nnU-Net leads on EF accuracy (MAE 6.10%, $r = 0.787$) and is 6.5× more parameter-efficient. The top four architectures (TransUNet, nnU-Net, UNet-ResNet, UNet-V1) form a statistically equivalent top-tier band. The pure-Transformer Swin-UNet is mid-tier on bulk Dice but degraded on EF, suggesting that its contours are subtly biased in a manner that compounds through Simpson's biplane integration.

The full-network HD95 of approximately 1.9 mm represents a substantial improvement over the originally reported CAMUS nnU-Net baseline of 4.3 mm. We attribute most of this improvement to training-protocol modernisation (cosine annealing, mixed precision, boundary loss) rather than to architectural changes. This observation is useful as a baseline calibration point for cross-paper comparison of CAMUS results.

Three directions remain open. (i) Cross-dataset transfer to EchoNet-Dynamic and ACDC would test whether the architectural ranking holds beyond CAMUS. (ii) 3D echocardiography requires architectures and pretraining recipes outside the scope of this benchmark. (iii) The recent wave of selective state space models (Mamba, Mamba-2, VMamba) is the subject of a separate, dedicated companion study (a companion study).

We release the complete training, evaluation, statistical-analysis, and figure-generation pipeline, together with the nine model checkpoints and per-patient JSON / CSV outputs that produced this paper, to lower the barrier to reproducing and extending these results.

Data and code availability

The CAMUS dataset is publicly available from `https://www.creatis.insa-lyon.fr/Challenge/camus/`. Training, evaluation, and analysis code; trained model checkpoints; and all intermediate JSON / CSV outputs supporting this paper are released at `https://github.com/TODO/repo`.

Acknowledgements

[TODO: Funding sources, compute, clinical reviewers, dataset providers.]

References

[1] Bao, H., Dong, L., Piao, S., Wei, F., 2023. SwinMAE: Self-supervised pretraining for vision transformers via masked image modelling. arXiv preprint .

[2] Bland, J.M., Altman, D.G., 1986. Statistical methods for assessing agreement between two methods of clinical measurement. The Lancet 327, 307–310.

[3] Cao, H., Wang, Y., Chen, J., Jiang, D., et al., 2022. Swin-Unet: Unet-like pure transformer for medical image segmentation, in: ECCV Workshops.

[4] Chen, J., Lu, Y., Yu, Q., et al., 2021. TransUNet: Transformers make strong encoders for medical image segmentation. arXiv preprint arXiv:2102.04306 .

[5] Chen, L.C., Zhu, Y., Papandreou, G., Schroff, F., Adam, H., 2018. Encoder-decoder with atrous separable convolution for semantic image segmentation, in: ECCV, pp. 801–818.

[6] Demšar, J., 2006. Statistical comparisons of classifiers over multiple data sets. Journal of Machine Learning Research 7, 1–30.

[7] Dosovitskiy, A., Beyer, L., Kolesnikov, A., et al., 2021. An image is worth 16x16 words: Transformers for image recognition at scale, in: ICLR.

[8] He, K., Zhang, X., Ren, S., Sun, J., 2016. Deep residual learning for image recognition, in: CVPR, pp. 770–778.

[9] Hu, J., Shen, L., Sun, G., 2018. Squeeze-and-excitation networks, in: CVPR, pp. 7132–7141.

[10] Huang, G., Liu, Z., van der Maaten, L., Weinberger, K.Q., 2017. Densely connected convolutional networks, in: CVPR, pp. 4700–4708.

[11] Isensee, F., Jaeger, P.F., Kohl, S.A.A., Petersen, J., Maier-Hein, K.H., 2021. nnU-Net: a self-configuring method for deep learning-based biomedical image segmentation. Nature Methods 18, 203–211. doi:10.1038/s41592-020-01008-z.

[12] Kervadec, H., Bouchtiba, J., Desrosiers, C., Granger, É., Dolz, J., Ben Ayed, I., 2019. Boundary loss for highly unbalanced segmentation, in: MIDL.

[13] Kolesnikov, A., Beyer, L., Zhai, X., et al., 2020. Big transfer (BiT): General visual representation learning. ECCV .

[14] Lang, R.M., Badano, L.P., Mor-Avi, V., et al., 2015. Recommendations for cardiac chamber quantification by echocardiography in adults. Journal of the American Society of Echocardiography 28, 1–39. doi:10.1016/j.echo.2014.10.003.

[15] Leclerc, S., Smistad, E., Pedrosa, J., Østvik, A., Cervenansky, F., Espinosa, F., Espeland, T., Berg, E.A.R., Jodoin, P.M., Grenier, T., Lartizien, C., D'hooge, J., Lovstakken, L., Bernard, O., 2019. Deep learning for segmentation using an open large-scale dataset in 2D echocardiography. IEEE Transactions on Medical Imaging 38, 2198–2210. doi:10.1109/TMI.2019.2900516.

[16] Lin, T.Y., Dollár, P., Girshick, R., He, K., Hariharan, B., Belongie, S., 2017. Feature pyramid networks for object detection, in: CVPR, pp. 2117–2125.

[17] Liu, Z., Hu, H., Lin, Y., et al., 2022. Swin Transformer V2: Scaling up capacity and resolution, in: CVPR.

[18] Liu, Z., Lin, Y., Cao, Y., et al., 2021. Swin Transformer: Hierarchical vision transformer using shifted windows, in: ICCV.

[19] Milletari, F., Navab, N., Ahmadi, S.A., 2016. V-Net: Fully convolutional neural networks for volumetric medical image segmentation, in: 3DV.

[20] Oktay, O., Schlemper, J., Folgoc, L.L., et al., 2018. Attention U-Net: Learning where to look for the pancreas, in: MIDL.

[21] Ouyang, D., He, B., Ghorbani, A., et al., 2020. Video-based AI for beat-to-beat assessment of cardiac function. Nature 580, 252–256. doi:10.1038/s41586-020-2145-8.

[22] Painchaud, N., Skandarani, Y., Judge, T., et al., 2022. Echocardiography segmentation with enforced temporal consistency. IEEE Transactions on Medical Imaging .

[23] Ronneberger, O., Fischer, P., Brox, T., 2015. U-Net: Convolutional networks for biomedical image segmentation, in: MICCAI, pp. 234–241. doi:10.1007/978-3-319-24574-4_28.

[24] Smistad, E., Østvik, A., Haugen, B., Lovstakken, L., 2021. 2d left ventricle segmentation using deep learning. IEEE Transactions on Ultrasonics, Ferroelectrics, and Frequency Control .